



TECHNICAL UNIVERSITY OF KENYA
FACULTY OF APPLIED SCIENCES AND TECHNOLOGY
SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY
END OF SEMESTER EXAMINATION 2024/2025 SERIES
THIRD YEAR EXAMINATIONS FOR THE DEGREE
BACHELOR OF TECHNOLOGY IN INFORMATION TECHNOLOGY/
BACHELOR OF TECHNOLOGY IN COMPUTER TECHNOLOGY/
BACHELOR OF TECHNOLOGY IN COMMUNICATION AND COMPUTER NETWORKS
ECSI-ECII 3103: STATISTICAL FINANCIAL AND ACCOUNTING SOFTWARE

TIME: 2 Hours

Instructions to candidates:

This paper consists of FIVE Questions.

Answer Question ONE [30 Marks] and any other TWO Questions [20 Marks Each].

Write your college number on the answer sheet.

This paper consists of 4 printed pages

Candidates should check the question paper to ascertain that all the pages are printed as indicated and that no questions are missing.

QUESTION ONE (30 MARKS) (COMPULSARY)

- a) Define the following terminology as they relate to statistical and accounting software (4 marks)
- Variable
 - kurtosis
 - OLAP
 - Statistics
- b) Define the REA model as used in accounting systems modeling describing the steps it involves (4 marks)
- c) Describe any two main data types available within R and give an example of each (2 marks)
- d) Describe the following operations as it related to R (3 marks)
- Make the matrix A equal to the one below
$$\begin{matrix} 1 & 2 & 4 & 4 \\ 3 & 5 & 3 & 4 \end{matrix}$$
 - Rename the columns as follows blue, green, yellow , red (2 marks)
 - Add 3 to each element in the matrix and call this matrix B (2 marks)
 - Add A and B (2 marks)
 - Include a 5th column equal to [1,2] to your matrix A and call the new matrix D (2 marks)
- e) Differentiate between data view and variable view as used in SPSS (4 marks)
- f) Write the equivalent python command that we can use to select rows 5 through 10 and the first third and fifth column from a data frame called staff Data. (5 marks)

QUESTION TWO (20 MARKS)

- a) Differentiate between the following terms
- Define Computed variable and control variable (4 marks)
 - Inferential and descriptive statistics (2 marks)
- b) List any four data types available within SPSS when declaring variables
- c) Give any three benefits of using graphics as compared to numerical values in statistical representations. (3 marks)
- d) Describe the following terms as they relate to QuickBooks (4 marks)
- Chart of accounts

- d) Describe the three fraud schemes according to Association of Certified Fraud Examiners
(3 marks)

QUESTION FIVE (20 MARKS)

- a) Write a command that allows us to create a plot of the students performance in mathematics versus programming from a dataset called "students" (4 marks)
- b) Differentiate between revenue transaction life cycle and expenditure transaction life cycle describing the steps in each (4 marks)
- c) Describe any four goals of security within A.I.S giving an example on how each could be enforced (4 marks)
- d) Given a dataset called staff, Import that dataset into python using relevant libraries, and use it to create a new data frame. (4 marks)
- e) Sort this dataframe in question [d] above you just created using a column called salary. (4 marks)